CARTESIAN MODULES AS DESCENT DATA ON A SIMPLICIAL SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: single-substitution sibling of lemma-characterize-cartesian (this batch, L46306); pairing note recorded.

Editorial note. The label lemma-characterize-cartesian-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Sheaf-level characterization. Same unit, Lemma lemma-characterize-cartesian. the full equivalence construction for cartesian sheaves of sets and abelian groups, transplanted to modules.

Source reference. Original reference 08N2: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2561–2561. Prerequisite pointer invoked at proof line(s) 2663; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2644; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$. The category of cartesian $\mathcal{O}$-modules is equivalent to the category of pairs $(\mathcal{F}, \alpha)$ where $\mathcal{F}$ is a $\mathcal{O}_0$-module and*

$$\alpha : (f_{\delta_1^1})^*\mathcal{F} \longrightarrow (f_{\delta_0^1})^*\mathcal{F}$$

is an isomorphism of $\mathcal{O}_1$-modules such that $(f_{\delta_1^2})^*\alpha = (f_{\delta_0^2})^*\alpha \circ (f_{\delta_2^2})^*\alpha$ as $\mathcal{O}_2$-module maps.

Proof. The proof is identical to the proof of Lemma 08N2 with pullback of sheaves of abelian groups replaced by pullback of modules. □

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